BunBau (資本寶) - Milestone v6.6 Implementation Plan
Universal Multi-Modal Ingestion Panel (PDF, CSV & Vision) & Context-Aware Tax Expert Ingestion Architecture

Document Version: `v1.0`
Target Milestone: `v6.6`
Application Version: `2.7.8`
Build Number: `Build 123`
Architecture Version: `v2.9.1 Universal Multi-Modal Tax & Portfolio Ingestion Architecture (PDF, CSV, Vision)`
Mandate Verification: *"You also need to update the application version/build/architecture version for every iteration build."*

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1. Problem Statement & Root Cause Analysis

Based on user review and visual analysis of the current UI (Screenshots 1 & 2):

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| SCREENSHOT 1 DEFECT: Ingestion Panel Mismatch for Tax Expert                                         |
|                                                                                                      |
| 1. Header is hardcoded to "Portfolio Ingestion" even when triggered from or for Tax Expert.          |
| 2. Top ribbon only has 2 tabs: [Brokerage CSV Export] [Screenshots (Vision)]. No PDF tab exists.     |
| 3. Dropping tax return PDFs into the CSV tab mislabels them as "CSV" with "Parsed Holdings (0)".     |
| 4. No strict per-tab file type enforcement (allows PDFs into CSV queue, creating confusion).         |
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| SCREENSHOT 2 DEFECT: Tax Uploader Modal Only Accepts PDFs                                            |
|                                                                                                      |
| 1. "Upload Tax Returns (IRS Form 1040 & Form 540)" strictly rejects any non-PDF file.                |
| 2. Users with 1099 CSVs, Schedule D trading CSVs, or scanned W-2/1040 screenshot images cannot       |
|    ingest them into Tax Expert.                                                                      |
|                                                                                                      |
| 3. Lacks universal multi-modal tabs and multi-format batch queuing.                                  |
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2. Architecture & Design Blueprint

```mermaid
flowchart TD
    subgraph TriggerContext ["Context Detection & Smart Ingestion Routing"]
        UserTopBar["Topbar [Upload Data] Button"] --> RouteCheck{"Active Tab Context?"}
        TaxBanner["Tax Demo / Live Banner [Upload]"] --> TaxDirect["context: 'tax'"]
        RouteCheck -- "activeTab === 'tax'" --> TaxDirect
        RouteCheck -- "Other tabs" --> PortDirect["context: 'portfolio'"]
    end
```

```
    subgraph UnifiedModal ["UploadValidationModal (Universal Ingestion Hub)"]
        TaxDirect --> SetTabPDF["defaultTab: 'pdf' | Title: 'Tax & Portfolio Ingestion'"]
        PortDirect --> SetTabCSV["defaultTab: 'csv' | Title: 'Portfolio Ingestion'"]

        SetTabPDF --> TabRibbon["Top Tab Ribbon: [
```